Research Assignment 2

The language of Business Analytics

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Course: Data Science
Due Date: 27 April 2026

1. Business Analytics

→ Is about using numbers and computers to look at data and find patterns and connections that helps companies make decisions.

eg. logistics firms analyze traffic data and delivery schedule to identify the most efficient, cost-effective shipping routes.

Source: Evans, J.R. Business Analytics.

2. Data Analytics

→ The process of taking raw data and turning it into actionable insights.

eg. Tik tok access specific video metrics, including views, likes, comments, and shares to understand content performance.

Source Jonathans. walker

3. Business Intelligence (BI)

→ Is the use of tools and systems that collects, process, and display business data in reports and dashboards. it helps organizations understand past and current performance

eg. By blending financial data with operations, marketing and sales data, users can pull insights from which decisions can be acted upon and understand factors that impact profit and loss.

Source: Andy Morris
Harvard Business Review

4. DATA-DRIVEN DECISION MAKING

→ It's about facts and real information, no assumptions. Businesses use data-driven decision making to be safer and get results based on facts rather than assumptions.

Source Mckinsey & company.

5. BIG DATA

→ It refers to large, and complex sets of data that traditional tools cannot process easily.

e.g. Social media platforms process millions of posts, likes, and comments every day.

Source: IBM / Investopedia

6. DESCRIPTIVE ANALYTICS

→ The process of using current and historical data to identify trends, patterns and relationship to understand what has happened in a business.

→ It provide context before attempting predictive or prescriptive analysis.

e.g. Sales Dashboards - visualizing current sales figures against targets, highlighting top-performing products or regions.

Source:

7. DIAGNOSTIC ANALYTICS

→ Is the process of examining data to determine why something has happened, by the roots of it by identifying causes and relationships in data.

eg:

e.g. Looking at aggregated data (e.g. total sales) and breaking it down by region, product, or time to find the root cause.
Source: Harvard Business Review

8. PREDICTIVE ANALYTICS
→ These are analysis that is based on predictions like predicting the outcomes of something.
e.g. When a bank predicts which customers are likely to apply for loans next month.
Source: Mckinsey Insights

9. PRESCRIPTIVE ANALYTICS
→ Suggests the best action to take based on predictions and data insights
e.g. An Online Store recommends the best discounts percentage to increase sales.
Source: IBM Analytics

10. DATA LITERACY
→ The ability to read, work with, analyze, and communicate with data to make informed decisions, rather than relying on intuition.
e.g. Marketing manager understands graphs and uses them to explain campaign performances.
Source: Harvard Business Review

SECTION 2: Data, Measurement & Reporting

11. STRUCTURED DATA

→ It is a data that is stored in table with rows and columns.

eg. Spreadsheet (Excel)

Source: When a company stores employees details such as ID number, name, department, and salary in an Excel sheet.

Source: Investopedia.

12. Unstructured Data

→ It is a data that does not have a consistant format.

eg. Gallery Folder in the phone

Source: IBM / McKinsey

13. DATA WAREHOUSE

→ Is a system used to store large amounts of historical data from multiple database for analysis and reporting.

eg. Shoprite combines sales data from all branches into one central warehouse for monthly reporting.

Source: Evans, JR. Business Analytics.

14. DATA MINING

→ Is the process of analyzing large database to discover patterns and useful information.

eg. A bank analyses customer transactions to identify unusual spending behaviour that may indicate fraud.

Source: Harvard Business Review

15. DATA VISUALIZATION

→ Is the presentation of data using visual tools such as Charts, graphs, and dashboards. It helps people understand complex information quickly.

e.g. A sales manager uses a bar chart to compare monthly sales across different regions.

Source: Microsoft power BI / Tableau

16. DASHBOARD

→ Is a graphical interface to view and analyze real-time data from sensors and devices to ~~mash~~ make informed decisions.

e.g. Unisa dashboard, displays login to myModules, Admin e.t.c.

Source: google Analytics.

17. METRIC

→ Is any measurable value used to track performance or activity. It can be any number that helps provide information.

e.g. The Number of daily website visitors is a metric

Source: Hubspot Academy.

18. Key Performance Indicator (KPI)

→ Is a specific type of metric that is directly linked to business goals and decision-making.

e.g. for an online Store, monthly sales revenue can be a KPI because it measures business success, while page views may simply be a metric

Source: Harvad Business Review

19. BENCHMARK

→ Is a standard, reference point, or criterion against which products, services, or performance metric are measured, compared, and judged.

e.g. A company compares it's 10% conversion rate against the industry average of 8%.

Source: Statisa

20. SEGMENTATION

→ The process of dividing a broad entity as a market, audience or data set, into smaller, distint subsets on shared characteristics

e.g. A clothing store separate customers into male, female, and children's shopping categories.

Source: google Analytics.

SECTION 3: Customer & Sales KPIs

21. Conversion Rate

→ Is the percentage of people who complete a desire action out of the total number of visitors or users. The action could be making a purchase, signining up or downloading something

e.g. If 1000 people visit an online store and 50 make a purchase, the conversion rate is 5%

Source: Hubspot

22 CUSTOMER ACQUISITION COST (CAC)

→ Is the total amount a business spends to gain one new customer. It includes marketing, advertising, sales, salaries and promotional costs.

eg. If a company spends R10 000 on advertising and gains 100 new customers, the CAC is R100 per customer.
Source: Harvard Business review.

23. CUSTOMER LIFETIME VALUE (CLV/LTV)
→ Is the total amount of profit or revenue a business expects to earn from a customer over the entire relationship
eg. If a customer spends R500 every month for 12 months, the CLV is R6 000.
Source: Mckinsey

24. CHURN RATE
→ Is the percentage of customers who stop using a company's product or service over a certain period.
eg. If 10 out of 100 subscribers cancel their membership in one month, the churn rate is 10%.
Source: Investopedia.

25. CUSTOMER RETENTION RATE
→ Measures the percentage of customers a business keeps over time. It is the opposite of churn rate
eg. If a company keeps 180 out of 200 customers in a month, the retention rate is 90%.
Source: google academy

26. NET PROMOTER SCORE (NPS)
→ Measures customer loyalty by asking how likely customers are to recommend the business to others, usually on a scale from 0 to 10.

eg. A resturants sends a survey ~~of~~ asking customers, "How likey are you to recommend us to a friend".
Source : Harvard Business Review

27. Average Order Value (AOV)
→ Is the average amount of money customers spend when they place an order
eg If total sales are R20 000 from 100 orders, the AOV is R200.
Source: google Analytics.

28. SALES FUNNEL
→ Is the journey customers follow from first becoming aware of a product to making a purchase. Common stages include awareness, interest, consideration, and decision.
eg. A customer sees an ad on facebook, visits the website, adds a products to the carts, and then buys it
Source: HubSpot

SECTION 4: MARKETING & CAMPAIGN ANALYTICS

29. MARKETING CAMPAIGN
→ Is a planned and coordinated set of activities used to promote a product, service, or brand. It may include social media, email, TV, radio etc
eg. A clothing store runs a winter sale campaign using Instagram ads, SMS promotions, and email newsletters.
Source: google ads

30. Post - Campaign Analysis

→ Is the process of reviewing and measuring the performance of a marketing campaign after it has ended. It helps businesses understand what worked well and what needs improvement.

e.g. After a black Friday promotion, a company analyses sales, clicks, and customer engagement.

Source: Harvard Business Review

31. Incremental Revenue

- Is the additional income generated directly because of a specific campaign, strategy, or action.

eg. If normal weekly sales are R50 000 and after a campaign sales rise to R70 000, the incremental revenue is R20 000.

Source: McKinsey

32. Return on Investment (ROI)

→ (ROI) measures how profitable an investment or campaign is compared to its costs.

$$\frac{\text{Profit} - \text{cost}}{\text{cost}} \times 100$$

eg. If a company spends R5 000 on a campaign and earns R8 000 profit, the ROI is 60%.

Source: Investopedia

33. Return on Ad Spend (ROAS)

→ Measures the amount of revenue earned for every rand spent on advertising.

$$\frac{\text{Revenue from Ads}}{\text{Advertising Cost}}$$

e.g. If R2 000 spent on ads generates R10 000 in Sales
the ROAS is 5:1
Source: google Ads

34. Click-Through Rate (CTR)
→ Is the percentage of people who click on an ad, email or webpage link after seeing it.

$$\frac{\text{clicks}}{\text{impressions}} \times 100$$

e.g. If 500 People see an ad and 25 click it, the CTR is 5%
Source: google Analytics

35. BOUNCE RATE
→ is the percentage of visitors who leave a website after viewing only one page without taking any action.
e.g. If 100 people visit a webpage and 40 leave immediately the bounce rate is 40%.
Source: google Analytics.

36. A/B Testing
→ Testing is the process of comparing two versions of webpage, ad, or email to determine which one performs better.
e.g. A company test two email Subject lines to see which one gets more opens.
Source: Hubspot Academy

37. LEAD GENERATION
→ is the process of attracting and collecting information from potential Customers who may be intersted in a Product or Service.

eg. A website asks visitors to enter their email address to download a free eBook
Source: Hubspot

38. Marketing Attribution
→ Is the process of identifying which marketing channel or campaign should receive credit for a sale or conversation.
eg. A customer first sees an ~~Tick~~ Tick Tok ad, later clicks a google ad, and then makes a purchase. Attribution helps decide which channel gets the credit.
Source: google Analytics / Hubspot.

Reference.

- Evans, J.R. Business Analytics
- Haward Business Review (HBR)
- Mckinsey & Company Insights
- Investopedia
- Google Analytics Help
- Hubspot Academy.